

Jack Ford

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EDUCATION

Duke University

Durham, NC

B.S.E in Electrical and Computer Engineering and B.S in Computer Science

Expected May 2030

- 1 of 12 selected from 60,000+ Duke applicants for full-ride through the Angier B. Duke Scholarship

The Haverford School

Haverford, PA

GPA: 4.3/4.0 | SAT: 1560/1600 | PSAT: 1520/1520

June 2026

- **Honors:** Hormell Foods 10 Under 20 Food Hero, National Merit Scholar, Congressional App Challenge Winner, Summa Cum Laude, University of Pennsylvania Book Award, Alumni Association Key Man Award

EXPERIENCE

Software Engineering Intern

Summer 2026

Forte3D LLC

Philadelphia, PA

- Interned at Forte3D, a Shark Tank company creating the world's first 3D-printed carbon fiber string instruments
- Trained custom Gaussian Process Regression model with Bayesian Optimization for printer and filament settings
- Reduced filament usage by 34.2% and improved cello durability by 16.5% for high humidity environments
- Redesigned storefront using Javascript and HTML/CSS to add custom UV-painted cello variant selection

Computer Vision Research Intern

Summer 2025

GRASP Laboratory, University of Pennsylvania

Philadelphia, PA

- Researched monocular visual inertial odometry for event camera-based optical flow under Prof. Kostas Daniilidis
- Developed a CNN-LSTM PyTorch pipeline to predict 6-DoF motion with learned uncertainty for EKF fusion
- Reduced driving localization error by 79.3% (ATE) and 47.3% (RTE) over state-of-the-art inertial baselines
- Authored a 5-page technical research paper and delivered a research presentation to 50+ students and teachers

Senior Co-President, Executive Director

June 2024 – June 2026

Food4Philly Inc.

Philadelphia, PA

- Coordinated 300+ volunteers across 20+ high schools as the primary leader of a nonprofit fighting food insecurity
- Implemented and led biweekly food distributions donating over 500,000 pounds of food to Hunting Park, PA
- Featured on Fox News, ABC News, the Philadelphia Inquirer, and regularly endorsed by local congressmen

PROJECTS

Astro | C++, Python, ROS2, Docker

May 2026

- Designed and built a 3D-printed quadruped robot driven by 12 ODrive motors and an onboard Jetson Orin Nano
- Developed ROS 2 software for gait scheduling, inverse kinematics, leg control, and CAN motor communication
- Implemented IMU-based stabilization and attitude control, enabling stable standing, trotting, turning, and tilting
- Achieved a top speed of 2.5 mph, max payload of 5 lbs, and stabilization up to 60°(vertical) and 45°(horizontal)

Chess Robot | C++, SDL2, Arduino

October 2024 – September 2025

- Built an autonomous chess-playing robot that detects human moves and responds on a physical chessboard
- Designed custom PCB integrating 64 magnetic reed sensors and 8 shift registers for piece and move detection
- Programmed control software integrating a custom chess engine, 3-axis gantry, sensor PCB, and machine GUI
- Achieved 2000+ ELO and 95% move accuracy, going undefeated across 30+ games against human opponents

Food4Philly Membership Platform | Javascript, Apps Script, Wix REST API

July 2025

- Built an automated membership system for Food4Philly to process and organize data across its growing chapters
- Integrated Google Forms, Sheets, and Apps Script to clean data, validate entries, and standardize chapter data
- Connected to the Wix REST API to automatically add new members to the nonprofit's internal emailing system

TECHNICAL SKILLS

Languages: C/C++, Python, Java, C#, Javascript/Typescript, SQLite, HTML/CSS

Frameworks: ROS2, SDL2, OpenCV, PyTorch, Numpy, Matplotlib, Scikit, React, Node.js, Flask, JUnit, Bootstrap

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse